

Effortless control through eye movements. Quantum Interface and EyeTech also unveiled technologies that could help intuitively control cars, gaming systems and other devices by simply moving the eyes. In a pre-CES interview, Jonathan Josephson, COO/CTO of Quantum Interface, commented, "Interacting with technology should be fun and easy. It should also be natural and intuitive. If you think about it, we look with our eyes to see what we want before we do anything else."

Together with EyeTech, they unveiled what they claim to be the fastest and most natural way to interact inside the vehicle. Quantum Automotive Head-Up Display (QHUD) is a solution that combines eye tracking and predictive technology for easier adjustment of controls while driving. The eye-tracker detects the direction of the eye movement, while user intent is predicted and confirmed with touch or voice.

Adjusting seats. Infotainment apart, another task that often distracts drivers is the pulling of levers to adjust seats. Scientists from Germany's Fraunhofer Institute for Silicate Research and Inringhausen GmbH have developed a seat that can be moved and adjusted using simple gestures. The prototype seat features a gesture interface located in a panel along its side. To begin adjusting, drivers activate the system by pressing on the panel's

synthetic covering, which veils a piezoelectric sensor. This kind of on-off prevents the system from accidentally reading random hand gestures as commands.

Once the system is powered on, instructions are read using proximity sensors that detect changes in the surrounding electrical fields caused by the driver's hand movements. Drivers can adjust the seat forwards or backwards, upwards or downwards, inclination of the thigh support or backrest simply by brushing fingers across the seat, which, in turn, activates the required motors. It is also possible to retrieve frequently-used settings by repeated pressing on the console. This is helpful in cases where a group of drivers use the same car. The researchers plan to commercialize the technology very soon.

Helping everyday Tom Cruises

John Underkoffler is the now-famous man who invented g-speak Spatial Operating Environment, the hands-free interface that Tom Cruise used in the iconic movie *Minority Report*, which always comes to mind when one thinks of gesture-based interfaces. Underkoffler then founded Oblong Industries to make real-world products using g-speak. Guess what? He figured that everyday workplace problem solving with Big Data is as challenging as the fantastic problems faced by the

actor in the movie.

The result is Mezzanine, an immersive collaboration tool that brings together physical environments, people, resources, much like what you witnessed in *Minority Report*. Mezzanine is based on Underkoffler's belief that a gestural computing system does more than just allow users to point and gesture at the computer to give instructions.

According to the company, a gestural computing environment is a spatial, networked, multi-user, multi-screen, multi-device computing environment. With Mezzanine, all walls turn into screens, pulling in people and workspaces from across the globe. A rack-mountable Mezzanine appliance acts as the hub for the collaborative work session. There is a 3D tracking system that handles in-room gestural-control aspects. This consists of a ceiling-mounted sensor array, a base radio and wireless wand-control devices.

The wand is a spatially-tracked handheld device that provides direct pointing control within Mezzanine. Apart from the many screens and cameras, there is a special whiteboard camera, which, coupled with the wand, enables easy snapshots of a traditional or digital whiteboard surface. Multiple-edge devices in and out of the room can be combined using a Web app. The resulting experience is apparently seamless, easy to configure and use, and quite close to a real-world collaborative experience.

Mezzanine makes use of g-speak, a spatial operating environment in which every graphical and input object has real-world spatial identity and position. This makes it possible for any number of users to move both 2D and 3D data around between any number of screens. g-speak is agnostic of screen size, which means a person working on a laptop in a café can work just as effectively as somebody participating from a conference room. Different components of this environment such as

Blinding light can be crippling, too

Gesture recognition under water is challenging. But, that is just one use case. Gesture recognition can be difficult in blinding light as well as foggy or smoky conditions, too. However, in many such situations it might also be the only possible mode of communication. Consider fire-fighting robots. Under choking environments, it might only be possible to communicate through gestures. What a dilemma!

Solutions like Episcan3D try to overcome this predicament. It is a low-power depth camera that can work under bright sunlight, scan difficult objects and image through fog. According to the team, "The sensor can capture images where most light from ambient sources is blocked out. It can also selectively image light paths based on type (single or multiple bounce) or based on the range of depths these correspond to."

Episcan3D was collaboratively developed by researchers at University of Toronto and Carnegie Mellon University. Of the four key researchers, two are Indian. Read more about it at www.csc.cmu.edu/~IJLIM/episcan3d/html/index.html#about